2 Ipseity and Language

What is alone true is that our open and personal existence rests on a primary basis of acquired and congealed existence.[1]

2.1 Traces of the other

How do we read another person's mind? The question regarding the comprehension of the other was first set, from a developmental standpoint, at the end of the 1970s in the domain of primatology. And it is in this domain of the natural sciences that, since then, this research area has gained ever increasing interest, even in a variety of disciplines that border upon primatology, such as developmental psychology, cognitive neurosciences, the philosophy of mind and psychopathology. The question set by Premack and Woodruff (1978) was, 'Does the chimpanzee have a theory of mind?' and the new field of study opened up by that question was baptized with the name of the question: 'theory of mind'.[2]

The problem of social comprehension, reformulated in the terms set by the theory of mind, runs thus: an organism has a theory of mind if this organism is capable of explaining the behaviours, or the actions, of others in terms of the mental states that generate those behaviours and those actions. Stated differently, one understands observed behaviour by theorizing about it. The ability to attribute mental states to others – which requires a capacity to produce metarepresentations – also provides the power of prediction. Put simply, if I can explain the behaviour of another organism in terms of desires, plans or intentions, I can also understand and anticipate that organism's subsequent course of action. The literature on deception and on social manipulation in the field of primatology has furnished arguments in support of the theory of mind, adding rhetorical force to scientific explanations (Goodall, 1971; deWaal, 1982; Byrne and Whiten, 1988).

On the other hand, the capacity to attribute mental states to others implies that the nonhuman primate is capable of conceiving of itself as possessing mental states and, more generically, is capable of recognizing itself as a Self. This presupposition underlies what can be defined as the argument by analogy: starting from (a) how I access my own mind and (b) the relationship between my mind and my body, I can infer that a body

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analogous to mine will be connected to a mind through modalities similar to mine. This is why I can start from another's behaviour and infer the mental states of that other related to such behaviour, either on the basis of innate mechanisms – TOMM[3] – or on the basis of theories – TT.[4] The studies which later provided strong support for this argument, and which enabled the problem of whether primates had a mind to be posed, concern mirror self-recognition. It was with a famous article by Gordon Gallup published in 1970 on the recognition of the Self by the chimpanzee that a rich line of research opened up which, almost 40 years later, still produces solid insights and reasons for reflection.

The most paradoxical aspect of this perspective is that, despite the fact that it developed in the field of evolutionary studies, instead of deriving man's mode of acquiring knowledge from the ape, it derives the ape's mode of comprehending from that of man. The ape is understood, that is, by deploying those very same categories of interiority which modern thought has adopted to understand man; it is attributed to those internal states which take shape in the closure of interior space.

But what happens if we try to change the question and ask ourselves how we encounter the other?

What happens, in other words, if we change the context of our research from the laboratories of primatology to everyday life, to the concrete situations of our daily existence? The problem then becomes: how do we understand the other immediately, when we meet him on the stairs, or at work or in the street – that is, before we attribute to him any mental states?

We have already analysed how oneself always lives ahead of itself, in a concerned absorption in the world, in a range of actions which goes from the simplest to highly intricate life plans. The experiential Self (oneself, namely everyone), encounters the other by way of how it feels, in the context of what it is doing, on the basis of what it needs, what it desires, what it expects or what it does not expect. On the other hand, we meet the other who is working, who is sad, who is in a great hurry, who is taking a walk, who is sitting in silence.

We do not encounter the other as men confined within their reciprocal interiorities – men who employ a theory of mind to attribute mental states to other men through an analogy with their own states – but in a world which we share with other human beings, in which we are already always open, on which we act and which acts on us. We meet the other first and foremost in his corporeity, in his style of conduct, in his actions and passions, which are made manifest in the environment of daily situations, in those contexts of ordinary, everyday life in which actions, passions or traces of the other are comprehended immediately without requiring any mental operations whatsoever.

In the wake of the discovery of mirror neurons (Gallese *et al.*, 1996; Rizzolatti *et al.*, 1996), many of those working in the neurosciences claim that this type of immediate comprehension of the other is founded on the automatic activation of a neural mechanism shared by the person who carries out a certain intentional act and the person who observes him doing so. That is, the act that is perceived elicits in the observer the activation of the same neural code as that of the act executed, thereby generating an unmediated resonance (Gallese, 2001, 2003) between the motor system of the person carrying out the action and the motor system of the observer. This mechanism, which Gallese calls 'embodied simulation', is claimed to underpin the comprehension of the meaning of action and the ascription of intentions.

What does all this mean in concrete terms? Let us return to the lamp on my table. It is getting dark, I am beginning to find it difficult to read, and without thinking about it I switch on the light. My body almost automatically performs a movement whose goal is to find the light switch in order to turn on the light. The action I perform is thus directed toward the achievement of some goal: it has an intention.

Now let us suppose that my wife comes into the study at that every moment and notes my action. How does my wife understand that I am seeking the switch in order to turn on the light? How does that movement manage to be meaningful to my wife? How does my wife manage to anticipate my intentions? Whether she wants to or not, my wife understands that I am about to switch on the light. Why?

2.2 Shared meaning

In order to furnish an answer to that question, Iacoboni *et al.* (2005) designed an experiment whose aim was to evaluate the importance of context in comprehending the intention of an agent. Twenty-three subjects observe three types of stimuli: (a) the action of grasping devoid of context, (b) only the context (that is, no action is taking place), (c) the action located in two different contexts – a table laid out for tea, suggesting the intention to drink, and a table after tea has been taken, suggesting the intention to clear the table.

Let us then suppose we are observing Mary taking hold of a teacup in a context suggesting a table about to be cleared. Through the mirror system I instantly recognize not only the action of taking hold of the teacup, but I also anticipate the immediate goal connected to the accomplishment of that action. Is it possible to say I have anticipated Mary's intention? Yes and no. Yes, because Mary really did want to take hold of the cup. No, because what meaning grasping the cup has for Mary remains obscure.

The data obtained through functional magnetic resonance imaging (fMRI) indicate that the mirror neurons of the premotor area appear to be involved in the attribution of intentions, in the sense of automatically anticipating the goal of a chain of actions, the connection between those actions evidently being suggested by the context. In the case of Mary taking a cup from a table laid out for serving tea, for example, the sequence anticipated is made up of that set of actions which comes to a conclusion with the carrying of the cup to Mary's mouth. It is as if the typicality of the context were automatically interpreted through the unconscious activation of a chain of motor acts connected to the discharge of neurons which are 'logically correlated'. Merleau-Ponty's (1962: 197) words return to mind: 'Already motility, in its pure state, is equipped with the power to attribute meaning'. That is, my body generates each time, and maintains, a meaning starting from its use in a space of shared actions.[5]

Does this explanation, which undoubtedly demonstrates the anticipation of the goal in a chain of actions in a typical context, really account for the comprehension of Mary's action? This poses a problem that does not concern the explanation as to why Mary took the cup (Jacob and Jeannerod, 2005). Mary actually took the cup in one case to clear the table and in the other case to drink. The question at issue is, rather, to understand why, on the one hand, Mary's action is manifestly comprehensible, so much so that we immediately understand its meaning and we anticipate its consequences, and why, on the other hand, what escapes us is the meaning that Mary assigns to her action. That is to say,

that while the meaning of the action is obvious, especially when the action is related to the context in which it was carried out, what remains hidden is the co-original link of the action with the person who carries out that action. This is the aspect about Mary that I cannot fathom. And perhaps this also corresponds to the limits to comprehending the other, as well as to the source of the regeneration of meaning.

In order to render those limits manifest, let us transfer the experimental situation into real life. Let us imagine that Mary has invited a friend, Linda, whose acquaintance she made only recently, to her country house for the weekend.

Let us suppose that Linda is still sitting at the table, because she was late in coming down to breakfast, while Mary begins clearing up. While she is conversing with Linda about this or that matter, Mary's face is calm, betraying absolutely no emotion. As soon as Mary makes to clear the table Linda immediately grasps her intentions. This means that, from the sensorimotor standpoint, Linda's mirror system (her mirror neurons generating shared activations) resonates with Mary's anticipating the goal of the chain of actions.

But that chain of actions, clearing away the table, can be understood by Linda as a gesture of courtesy in her regard, or as a sign of impatience at her having arrived late for breakfast, or it may be attributed to Mary's routine habits, or it may even pass unobserved.

On the other hand, Mary may want to clear up because she likes a tidy kitchen, or because she has nothing better to do, or perhaps because she wants to take Linda out on a trip, or because she wants to make Linda feel at home. That is to say, that action, whose meaning is shared inasmuch as it is carried out by Mary and simulated by Linda, is concurrently personal to the who that carries that action out and to the who that simulates it. And it is precisely this second aspect, the mineness of the action in terms of significance, which opens a hiatus of meaning between Mary and Linda, between me and the other – as a result of which the other may surprise me, he may disorient me, he may make me think! 'If the other is really another,' writes Merleau-Ponty (1962: 198), 'then at a certain point I will be surprised, disoriented, and we will meet again, no longer with what we have that is similar, but with what we have that is different, and that presupposes a transformation of myself and of the other too: our differences will have to cease being opaque, they must become meaningful'.

Vittorio Gallese (2005: 43) contends that 'an important component of this similarity resides in the common experience of action'; nevertheless, a further equally important component resides in the difference in the way the shared experience has been perceived by each single individual. This begs the question of how and why experiencing one and the same act can give rise to a range of meanings which, for instance, in the case of Linda may vary from pleasure to indifference. We will explore this matter in greater depth in the following chapter.

Let us now, instead, analyse the immediately comprehensible aspect of action, that part that may be called public, almost objective, so to speak: we will call it the intentional content of the action. Mary begins to clear the table. Linda perceives Mary's action and, after about 30–100 ms, her motor system begins to tune in to the action she has perceived Mary carrying out. Linda's motor system is reflected in Mary's motor system through the simulation of the action observed. At the neural level, this corresponds to the activation of a series of mirror-matching neural circuits. Nevertheless, the activation of Linda's mirror neuron system (MNS) simulates the intention expressed in Mary's action, not Mary's being mine of the action. We could say that the MNS simulates the intentional

content of the action, the action as such, but not the agent that accomplishes the action, not the who that acts.[6]

In fact, resonance systems are activated both by one's own actions and by observations of those actions accomplished by others, constituting a sort of neutral (that is, neither first-person nor third-person) common code for perception and action. 'As soon as we see someone carrying out an act or a chain of actions,' write Rizzolatti and Sinigaglia (2006: 127), 'then whether that someone likes it or not, his movements immediately acquire significance for us. Obviously, the same applies to the opposite situation: every one of our actions immediately takes on meaning for those observing those actions. Possession of the MNS and the selectivity of their response thus determine a space of shared action, within which each act, each chain of actions, whether they be executed by us or by others, immediately appear to be inscribed and comprehended without such operations requiring an explicit and deliberate cognitive operation' (Gallese, 2005; see also Hurley, 2006). Stated differently, the experiential Self which perceives an action carried out by a conspecific is actively reflected in the action perceived, thereby generating a similar action (or 'simulation').[7]

Nevertheless, whoever simulates action by means of this process of reflection in the other, concurrently experiences his own mode of comprehending that action: this may range from indifference to active and meticulous participation, to complete absorption. This means that while Linda's motor system comprehends Mary's action by simulating it, that action may go unnoticed by Linda herself, or it may awaken her interest, or it may even become the sole focus of her attention. The being mine of the act corresponds to the personal sphere of experience. It does not need to be internal in order to be private.[8]

From this perspective, the activation of the MNS helps explain how the Self attributes meaning to its encounter with the other, despite the fact that this encounter does not exhaust the comprehension of alterity. These systems are probably part of the neural substratum of that social being which is constitutive of being oneself. 'Constitutive' is not intended here only in its oblique sense, as in Heidegger, namely that we already always encounter the other through things, through artefacts, as if objects alluded to their users, their traders. It is intended, instead, as sociality experienced directly, face on, without it being dazzled by the reflection of the horizon, without leaning on the context for support, inasmuch as it is only through the sharing of meaningful acts anchored in our living bodies that the how I grasp myself through such sharing, the how I reflect myself in it, emerges to experience concurrently with the comprehension of the other. Such sharing is constitutive of being oneself; comprehending oneself is contemporaneously comprehending others.

Unlike the solipsism, where the other corresponds to a datum furnished by reflection, in this perspective we encounter the other starting from his being with us in the world, as if the other were, at one and the same time, text and context, image and background. This is evident right from the first few hours of a baby's life.

2.3 *Finding* oneself in the world: suggestions from phenomenology

'For unto us a child is born' (Isaiah 9:6). With these simple and powerful words, the Holy Bible announces the 'glad tidings'.[9]

A new life enters the world, a voice never heard before breaks into shared space, disrupting it; the face of another, who has never appeared before, stages its entry into existence. It is the epiphany of the face, a presence which shines with its own light, which signifies by way of itself, exteriority which does not require a context, evidence which instantly expresses itself. 'The face speaks. The manifestation of the face is already discourse' (Levinas, 1961: 64).

When we think of childhood, the inexhaustible theme of Levinas' work is blatantly clear. Not the face of the widow, the orphan, the stranger, but a face which is even more naked, that of a son of man, whose naked gaze conveys vulnerability and supplication, expression and injunction, manifestation and commandment.

Levinas' strong words befit this face, almost as if the insistent use of the trope of hyperbole no longer corresponds to the systematic practice of excess – which nevertheless is a major feature of Levinas' style of philosophical argumentation – but constitutes the only adequate measure. As if there really were no other means bar hyperbole to describe a child's bursting into the world at birth, his 'subversive' entry, which is independent of any significance.

It is the entry of that body into the world, its visible vulnerability, its absolute alterity, that moves me into taking on responsibility. Even more important, his fragility questions my being to the very last of the cells of my flesh, it obsesses me, it haunts me, it expels me from myself, it makes a hostage of me, 'it inflicts on me a wound of love' – only the exaggerated words of Levinas can explain that manifestation which is so acute and so serious, and which represents the 'expressive violence' brought about by the fragility of a child's body. His being alive demands my time, my care, my dedication, my concern . . . my desire. I have to make room for him in the world where I exist, I have to build a warm home, one which is safe, I have to protect him growing up, I have to provide him with everything that is necessary for him to develop his own story. His being in life calls on me to respond, his body is the commandment which obliges me to take on responsibility.

My son's injunction, which is made even more 'visible' by my son 'breaking into' my existence, corresponds to being affected by him, and my being affected by him constitutes the manifestation of that passivity which is specific to ipseity.

'In actual fact, at the phenomenological level,' says Ricoeur (1995: 432), 'the many modes in which the other affects the comprehension of the Self for itself indicate, to be precise, the difference between the ego which asserts itself and the Self which recognizes itself only through the ways in which it has been affected'. And in actual fact, ipseity is characterized by openness and by its revelatory function and is, therefore, very different from an I falling back on its own interiority and separate from the world and from others. It is by virtue of this constitutive trait of openness, which is structured according to the various modalities in which ipseity *finds* itself in the world and with others, that alterity is an integral part of ipseity. In the structure of ipseity itself, a structure which is continually decentred and open to the other, action and passion intertwine, as do feeling and being affected, accepting and being called upon, giving and receiving, listening and speaking or, as Ricoeur puts it, attesting and being injuncted. Thus, alterity, which from a phenomenological standpoint is matched by the experience of the passivity of oneself affected by the other, is constitutive of ipseity: it is part of ipseity, as we shall see, in different ways, each of which has different origins and plays its part to different degrees. Passivity and agency are inseparably intertwined at the heart of intersubjectivity.

2.4 Body-to-body

The dialectics of action and of being affected that permeates the adult's interaction with the child is present and obvious from the very start, and it occurs with different modalities and on different levels.

Right from birth it is the body-to-body relationship with the caregiver that constitutes the privileged site of communication and, more generally, of the synchronization of cycles of intimacy which correspond, from a physiological viewpoint, to phases of mutual activation and deactivation. It is in the body proper that openness to the other is based. It is in inter-corporeal space that the child encounters and tunes in to the other, thereby achieving mutual comprehension. On account of this 'tuning in', the adult offers himself, not only as a source of caregiving, but also as a partner of combined resonance, as the source of consent and shared meaning. For the child, the caregiver thus represents the opportunity for him to contemporaneously gain access both to oneself and to shared human meaning. Hence, right from birth, understanding the other also involves understanding oneself through the meaningful signs that irradiate from the child's body to meet the other man in a common world.

The gradual constitution of one's mind, like that of other minds, thus becomes possible only through the combined construction of 'sensed' events, through the sharing of 'regulated' expressive states, which happen (affections) or which are produced (actions) initially by the body-to-body interaction with the caregiver. It is initially in this context that the child externalizes the rudiments of sense by coordinating his actions and passions with those of his caregiver.

The structuring of intersubjective processes thus goes well beyond the behavioural and motivational aspects to which the classic studies on attachment refer. Intersubjectivity thus assumes an ontological importance that can hardly be reduced to instinctive behaviours triggered by parental care. 'The expressive behaviours in affectionate chat and play,' write Trevarthen and Aitken (2001: 6), 'have no immediate role in the regulation of the neonate's physiological state, comfort or survival. They are distinct from maternal breast-feeding, stroking, holding, rocking, vocal comforting and the like'. That is to say, communication cannot be explained exclusively by instinctual behaviour or biological needs. Rather, it takes shape by way of the very existence of the child. Coming into the world corresponds, first and foremost, to openness to the being of the other . . . to being with the other.[10]

This openness is based on the child's body proper and comes about thanks to the capacity of the two bodies – one adult, the other infant – to coordinate their reciprocal comprehension. In other words, the expression produced by one body must count as a sign for another body, so that the perceiver may take his bearings, just as raising one's hand in a classroom indicates to the others the actor's desire to come into the discussion. This process leads to the interactants becoming familiar with a shared 'proto-language', a process which in its turn corresponds to the sedimentation of a communicative code and, at the same time, to the development of an inclination for a certain mode of feeling. Evidence of this is the so-called 'fear of strangers'. At about the age of six months, when a child manifests fear in the presence of an unknown person, what he is actually expressing is the distress created by the fact that the coordination constructed with his caregiver, which normally enables him to feel situated, 'doesn't work' with the stranger. Like an alien, the stranger is unable to coordinate his own expressions in the sequences

and in the contexts which the child recognizes as being shared and understood (Trevarthen, 2004).

The first synchronization of exchanges – which Locke (1993) argues could be considered as being a developmental precursor of vocal turn-taking – which begins to emerge during breastfeeding as a contingent alternation of the caregiver's jiggling and the infant's sucking, clearly highlights the fact that this capacity for social interaction is bred of bodily mediation. That communication as tuning – a process which presupposes that the adult possesses the sensibility to respond to the appeals made by the neonate – becomes even more evident from the first vocal turn-taking exchanges, which begin at about the seventh to eighth week of life (Papousek and Papousek, 1989). The quality of these proto-conversations depends on the contingency of the response provided by the caregiver, since from the very outset of language development it appears obvious that communication refers to the contingency of the position of the hearer and that of the speaker, to the way they find they are related to each other.

For example, in a study of 48 infant males, Masataka (1993) shows that 'the child in the absence of response changes flexibly the intervals between two consecutive vocalizations according to his recent experience of turn-taking with the adult'; that is, in relation to a pattern exhibiting communicative harmony which was developed in the relationship with the caregiver. Reciprocal adjustment, the brunt of which is borne by the caregiver,[11] allows the child to reflect himself by means of a human source of reference (who reflects himself in the child) and to grasp himself through the continuous flow of experience. This joint participation is already present during gestation (Fernald, 2004; Fifer, Monk and Grose-Fifer, 2004).

Moreover, it also seems that the passage that occurs around the age of three months, from a production of 'vocalic sounds' with greater nasal resonance to one of 'syllabic sounds' with greater oral resonance, is facilitated by the caregiver's vocal stimulation contingent to the infant's vocal production. At this age, however, for speech sounds to make vocal production easier, they must be quite familiar to the child (Bloom, 1988; Masataka, 1993, 1995). At the root of the special role played by contingency in the early stages of infant vocal behaviour there may well lie a sub-personal mechanism (in addition to other mechanisms), which allows the child – during a calm and enjoyable interaction – to translate the caregiver's vocal and facial expressions of emotional feelings into the infant's own similar bodily gestures.

It would thus appear that proto-linguistic production and reception imply a matching system which enables the child to reflect himself not only in the vocal gesture but also in the facial expression and emotional state of his partner (Warren *et al.*, 2006). The MNS might carry out that function, thereby enabling the construction of an intersubjective code for the comprehension of expressions (Gallese, 2007). The link between perception and production guaranteed by the mirror neurons (see below) could account for both the infant's capacity to imitate sound patterns (Kuhl and Meltzoff, 1996) and his ability to link the mouth movements he sees to the auditory signals he hears (Kuhl and Meltzoff, 1982; Kuhl, 2000).

Seen in this light, even the fundamental phenomenon of imitation takes on a complex meaning. If imitation is considered as a function of interpersonal communication – that is, within a context of recurrent interaction, as constituting a web of shared actions and emotions with the caregivers – then imitational behaviour is inscribed within the sphere of meaningful reciprocal behaviour, which obviously changes over time. The imitational

response thus becomes a mode of constructing or accessing a common code, a shared space of actions and emotions which constitute the foundation for the relationship (Gallese, 2005; Decety and Chaminade, 2005), but it concurrently acts as a mode of accessing oneself. And it is precisely the consensual status of imitational response that enables it, at its most advanced stages of development, to act as 'affirmation, acceptance and commentary' in the flow of interactions when the interlocutor provides a more emphatic ostentation of expressiveness. Imitation, in fact, implies the comprehension both of what another individual is doing and of the contexts in which such knowledge is put to use, namely those contexts in which such acts can be replicated (Rizzolatti, 2005). It is by virtue of shared meaning that the imitational act may be attributed a personal meaning, depending on the various environments in which the interlocutors may find themselves, the various contexts in which those acts may be put to use. Or else, the imitational response may be inhibited, as is demonstrated by a recent study by Meltzoff (2007) on the relationship between the child's anticipation of the adult's anger and possible imitative behaviour.

Imitation is thus not simply a means to acquire a certain behaviour pattern (a learning mechanism), but enables the child to regulate those processes of interpersonal involvement which express the different fields of individual motivation (Trevarthen, Kokkinaki and Fiamenghi, 1999).

Both expressive and vocal reciprocal imitation seem to play a very clear role in the use caregivers make of that singular way in which the child talks, often called 'motherese' or baby talk. This infant-directed speech, which has syntactic, semantic and prosodic features that are simpler than speech directed to adults, appears to be specifically designed to facilitate infant imitation (Ferguson, 1964; Kuhl *et al.*, 1997; Fernald, 1984); that is, motherese is based on the spontaneous vocalizations uttered by the infant which the adult matches and models, thereby generating an imitative form that serves as a contingency for infant vocal production (Masataka, 1993).

The typical higher pitch, the better articulation and the slower tempo of infant-directed speech, accompanied by exaggerated facial expressions and bodily movements, make the phonetic units of their native language easier for children to discriminate. On the other hand, exposure to a specific language – their native language – influences infants' phonetic perception, hence their sound production (Kuhl *et al.*, 1992). Recent studies of three-month-old infants deploying fMRI indicate that exposure to one's native language underpins the activation of left-lateralized brain regions comparable to those of adults, including the superior temporal and angular gyri and inferior frontal regions (Broca's area) (Dehaene-Lambertz, Dehaene and Hertz-Pannier, 2002).

The course of a 'felicitous proto-communication' is thus characterized by cooperative communication, which manifests itself in synchronic phrasing in which expressions of one interactant are reflected in those of the other interactant (synchronization), intertwined with phrasing in which the interaction is characterized by each partner alternately producing linguistic expression (alternation). A simple sequence could begin, for instance, with a vocalization expressing joy on the part of the child, to which the caregiver responds first with an expression of concentration, followed immediately afterwards by an amplification of the expression of joy, both at the visual and at the vocal level, with the child then overlapping with a further manifestation of joy accompanied by the clapping of his hands, reaching a sort of climax, to then return to a state of relaxation (Trevarthen, 1979).

2.5 The significativity of expressions and objects

The comprehension of an expression (such as a gesture) implies that the expression be interpreted as significant. But what is the significativity of the expression observed for the child? What does it consist of? And how is that meaning grasped?

We have already seen that one of the most important aspects of communication is that expressions are comprehended through a system of resonances (mirror neuron systems), which enables the child to reflect himself in the acts of the caregiver, thereby grasping himself. We have also underlined the fact that imitation – which implies the activation of the MNS (Rizzolatti, 2005) – has to be considered in the wider context of interpersonal communication, where the central role is played by the contingency of the position of the caregiver and of the infant, inasmuch as communication refers on each and every occasion to the combined situation of who is expressing himself and who is interpreting that expression.

Hence, the range of communicative responses seems to extend between two polarities: one in which one can perceive oneself by means of refraction through the other person's conduct, as in forms of imitation and in interactions expressing joy – which are often combined with non-basic emotions[12] (Trevarthen and Aitken, 2001) – the other in which the infant perceives himself through an activation which frees him completely from sharing, as, for instance, in the case of the triggering of a basic emotion.[13] For example, an episode of anger on the part of the caregiver may generate fear in the infant in as much as it activates a complex, coordinated set of bodily changes, without the activation of the child's fear being mediated by a mutual resonance system with the caregiver. Between these two polar extremes, a set of intermediate stages may be hypothesized which require coordination between the system of basic emotions, that of non-basic emotions and the perceptuo-motor resonance systems.[14]

From the fourth month on, the issue of the significativity of the expression undergoes an important transformation. The child, whose interaction up to that point was limited basically to a body-to-body relationship, begins to exert an initiative directed at the world of things, gradually acquiring control over the use of his arms and hands to reach and grasp objects. At the same time he follows the shifting gaze of the other person manipulating or seeking an object (Muir and Hains, 1999). In other words, the child perceives not only the adult's attention – of which he is emotionally aware – but also the objects at which the adult's attention is directed. This process of opening himself to the world represents a new stage in the development of the child, his progressive capacity to act in a consensual manner (which will reach its apotheosis at approximately nine months in the renowned 'joint attentional frame', to be discussed shortly). This change goes hand in hand with a new stage in the relationship with his caregivers. This transformation is particularly evident in the evolution of the types of game played: person–child games are supplanted by person–child–object games. However, since, at this age, the child is unable to combine acts which orient to the person through the use of objects, he cannot, for instance, connect the fact that a person is observing an object to the fact that that person intends to do something, for instance use the object. It is as if the child's interest in manipulating the object cannot be assimilated to the person.

On the one hand, the child begins to act with objects in an autonomous fashion: his coordination improves continuously and he learns to recognize the progress made in, and the consequences of, his course of action. On the other hand, the caregiver becomes his

first teacher and playmate, creating, participating in or varying – in the more advanced phases (seven to eight months) – the consequences of the possible outcomes stemming from the sequences of actions that the child undertakes during the game. Stated differently, the dimension of increasing autonomy, by means of which the child extends his own course of actions and, consequently, the spatial dimension of his world, is regulated by the caregiver's capacity to coordinate his own actions and emotions in a way that is contingent with those of the child.

A six-month-old infant can share with his caregivers a concatenation of routine actions, which interweave with emotions and with communications lasting many seconds. His increasing ability to sequentialize becomes even more transparent in his ability to grasp the temporal development of lullabies, an ability manifested by children of five to six months in different cultures. The musical structure of lullabies is roughly comparable to a classic narrative structure, with a regular rhythmical beginning, an apex that is emphatic to a greater or lesser degree, and a coda. This cycle is repeated several times in the course of the lullaby, and the child seems fascinated by the 'process of prediction and recognition which is usually described as cognitive, but which in this case is certainly accompanied by an emotional evaluation which, in parallel fashion, varies in a predictable fashion' (Trevarthen, 1977: 243).

The importance of the prosodic aspects of language in early infancy are supported by a study (Homae *et al.*, 2006) using infrared spectroscopy in three-month-old infants, which shows that prosodic information is mainly processed by the right hemisphere. Another event-related brain potential (ERP) study (Pannekamp, Weber and Friederici, 2006) indicates that in eight-month-old infants prosodic cues are relevant for sentence structuring.

2.6 Referential communication

The ability to order events sequentially and to predict the goal of a concatenation of events brings about an important change in communication. From approximately the age of nine months, the child is able to combine his sensorimotor and attentional capacities with those of the caregiver, enabling him to enter into a new type of communication characterized by attention-sharing – which is emotionally situated – on an event or on an object. Communicative pointing (but also showing, offering, giving), which often appears at this age, and which is restricted almost exclusively to third-person reference, is an unequivocal demonstration of this (Bates, Camaioni and Volterra, 1975). Joint attention may in fact be such because, in the course of the communicative event, the child can anticipate the adult's possible intention with regard to the action the latter will undertake involving the object. For instance, in pointing to a particular toy (in a given emotional context), the child predicts the possibility that the adult will offer him the toy, or that he will begin a game involving that toy. This implies that the child can understand that a communicative gesture involving an object can orient the goal of the caregiver's action, and vice-versa.

Such comprehension is facilitated by the reproduction on the part of the child of the adult's novel intentional actions on outside objects – as may be inferred from role-reversal imitation (Tomasello, 2003). For example, in the context of a cooperative game involving a toy, after having observed the goal-directed concatenation of actions executed by the

adult, combined with contemporaneous expressions of joy, equally harmonious imitation on the part of the child reverses the child–caregiver role relationship (Trevarthen and Hubley, 1978).

The joint attentional frame, which is emotionally situated, and which characterizes the onset of referential communication, creates a condition which multiplies the possibilities the child has of accessing the world to an unimaginable extent. Indeed, the manipulation of objects, just like combined focalization on shared domains of relevance in the course of interactions, enables the child to develop a common space of action, of emotions, of intentions in which he can *grasp* things, situations and behaviours in a new manner: as meaningful. As Gallese (2005) states, 'This space is *we-centric*'.

And in fact, 'from soon after their first birthdays, infants cannot help perceiving Daddy as "trying to clean the table" or "trying to open the drawer" – not simply as making specific bodily motions or producing salient changes of state in the environment – and these intentional actions are what they attempt to reproduce' (Tomasello, 2003: 27). What is more, these intentional actions begin to have a meaning which the child is capable of understanding because he is able to predict the forthcoming goals. If, for instance, an adult comes into the room with the pram, a joint attentional frame emerges that is focused on the (shared) anticipation of going for a walk. That is, the child comprehends the object on which the attention of both interactants is focused in the light of the prediction of a sequence of actions and in the sphere of the emotional context in which the object – as it has been understood – is embedded. Joint attention is possible because grasping adults' intentions corresponds to the infant's emotively situated anticipation of the consequences of actions. Against this common background the pram acquires shared meaning.

In fact, some of the earliest uses of object words refer to the entire organized sequence of actions and emotions in which the object is embedded. Thus the child uttering the single word 'pram' (accompanied by smiles and bodily movements) indicates that the child is producing an expression such as 'Good, it's time for a walk'.

Hence, for an object to become significant, the necessary condition is that the child acquire the ability to comprehend objects and situations starting from possible contexts of actions and emotions. Meaning is, so to speak, connected to a series of anaphoric references scaffolded and sedimented through the child's history of interactions with his caregivers, which constitute the contexts through which objects, situations and people become relevant for the child. This set of references, which the child does not comprehend explicitly and expressly, is the presupposition which enables words to be linked to meanings for the child. As is emphasized by Catherine Nelson (Nelson, 1996: 106), 'from about 12 months to about 36 months the language system is being developed within and in service to communicative contexts already established as social activities'. It is only on the basis of this relationship that the word (which at 12 months has the same value as pointing) can be understood, inasmuch as it indicates something – which does not yet appear in the word – on which speaker and hearer can agree.

Hence the close connection between joint attention and child language acquisition should come as no surprise (Bruner, 1983a, 1983b, 1978; Scaife and Bruner, 1975); nor should the amount of time infants spend in joint engagement with caregivers and the size of infants' vocabulary (Tomasello and Farrar, 1986; Smith, Adamson and Bakeman, 1988), nor the relationship between faster rates of infant vocabulary acquisition and more cooperative parenting (Akhtar, Dunham and Dunham, 1991; Baldwin, 1995).

The emergence of the child's 'first words', which at this age are typically produced in highly specific contexts or are event-bound (Barrett, 1986), is therefore founded on the comprehension of concatenations of actions and emotions occurring in shared everyday situations.[15] Recurring sequences are made up of organized, synchronic patterns which intertwine with the actions and emotions of the child and his caregivers. They are accompanied by signals that the adult repeats over and over again, marking the beginning, the development and the conclusion of the specific pattern. For instance, nappy changing, bathing, getting ready to go out, mammy or daddy coming back into the room, a visit from granny and so on; and even more complex games with objects involving sequences, cooperative activities accompanied by instructions, or statements, or emphatic expressions on the part of the mother, role reversal imitation and so forth.

When the child enters his second year of life, the type of object he wishes to play with gradually changes. Attention is progressively shifted on to the manipulation of things which are part of routine actions in daily family life (telephones, cups, spoons and so on). It is in the context of these situations pertaining to everyday existence that the first words acquire a referent. Words come into existence starting from what the child and adult can mutually agree upon. The comprehension of the child's first words, hence the development of early word meanings, is based on these shared practices which provide the child with the experiential, and also cultural, data that words refer to.

This origin, which hints at the fact that language is grounded in action and emotion, has been demonstrated by a series of studies of adult subjects (Pulvermuller, 2001, 2002).

Studies on the acquisition of personal pronouns (Bates, 1990) provide a complementary observation point on this developmental period. For example, from the onset of proto-linguistic communication up to the first stages of language, communicative pointing and early words are limited to third-person reference. That is to say that in the course of a communicative exchange, the child never points to himself or to the hearer (either with his index finger or by means of a personal pronoun), but to events or objects in the external world. He constitutes the subject, not the object, of his referring activity. This inability to grasp himself reflectively also manifests itself at the level of imitation.

It appears, that is, that before linguistic development has been completed, the child is capable of imitating action only if he refers the sequence to be repeated to himself, reproducing that sequence, becoming the active subject (for instance, feeding-himself, combing-himself). In other words, he repeats sequences of perceived events, and until the age of 18 months the referent of this imitation is himself (Trevarthen and Logotheti, 1989), while, instead, he is not able to grasp himself reflexively as agent. From 18 months on, children begin to combine words into rough sentences that allow them to symbolize highly repetitive scenes in their lives. The most frequent contents of these first utterances concern possession ('my teddy'), location ('car garage'), nonexistence ('no more soup'), recurrence ('tickle again') and actions ('me fall', 'car go') (Camaioni, 2001).

Together with the emergence of the child's first words and a spurt in his vocabulary, the period 18–24 months sees the child begin to recognize himself as the source of his own (shared) actions and emotions, as is shown by the classic studies on 'Me in the mirror recognition' (Berenthal and Fisher, 1978; Johnson, 1983; Lewis and Brooks-Gunn, 1979; Anderson, 1984; Damon and Hart, 1988; Butterworth, 1990).[16] At the level of language, this ability manifests itself in the onset of the use of an identifying referent, which is generally the personal pronoun 'I', or the child's own name, or the co-existence for a certain period of the name and the 'I'.

This new stage of development sets important questions.

First, is there a relationship between the construction of the sentence – which integrates into one single unit the identifying referent (the subject) and the predicating function (the predicate) – and the ability the child has just developed to recognize the reflection in the mirror as his own image? Second, what has generated this new ability to use the predicate (and therefore the verb) within a sentence, namely, of grasping time through the use of language?

2.7 Oneself in the mirror and in the refraction of language

After the child's second birthday, when the caregiver walks into his room and says, 'Ok, let's go to the park', the child understands that those words refer to a concatenation of actions, to an event that will take place shortly afterwards. That event, tinged by shared emotions, is located in a framework jointly built and sedimented over time. But while at 12–18 months the child employed his first words – which appeared to be a vocal gesture connected more or less rigidly to a given event – to recognize himself in a sequence of actions, predicting the sequence and concurrently recognizing its sense, once the child has entered language he 'interprets' the linguistic expression. The foundation of sense conveyed by the expression (which always disappears by dint of being hidden by the discourse) consists precisely in the shared aim toward a concatenation of actions and emotions connected to a background, and this convergence is produced through the joint act of who speaks and who listens.

In order for the child to comprehend the meaning conveyed by the linguistic expression, the words must first be perceived as being significant. That is to say, the child must grasp the expression as constituting a reference to a domain of shared actions and emotions, without it being directly, pragmatically, context-bound. Putting language to use gradually frees the word from the concrete context of use. This is the principal aspect dominating the pre-school years, which culminates in the development of narrative skills. We may state that by way of its use, language employs the voice and the hands (sign language) to reconfigure a sense that was already shared (*viz*. manifest) at a pre-reflective level in the sequence of actions and emotions that occur in the communicative exchange. In other words, it reconstructs the concatenation of events through sound.[17]

Linguistic communication is thus based from the outset – where undoubtedly such roots are more clearly discernible – on an antepredicative dimension which we do not understand explicitly, and which varies continuously for each one of us and on each and every occasion, depending on the circumstances, the richness of the content and its transparency. This is the context in which our actions and passions take shape. Against this background, what the discourse refers to stands out clearly and manifestly. The salience of this or that entity, of this or that situation, of this action or of that emotion which the linguistic expression refers to, depends, that is, on the pre-reflective context which does not appear in the discourse. It is precisely that context which furnishes the opportunity for the making of the assertion.

The other peculiar aspect consists in the fact that the comprehension of the sentence – whose content concerns a situation extant in the world which the speaker is referring to – comes about through the hearer giving it meaning which is, so to speak, private, in as much as it refers to the hearer's own experience of being-in-the-world. That is, while the

hearer is grasping what the sentence means, he relates the experience which the discourse is addressing to his own perspective. In so doing he appropriates himself, he includes and understands himself in one way or another according to the directions indicated by the discourse. The process engaged in by the listener is specular to that engaged in by the speaker: just as what the sentence means refers to what the individual subject who produces the utterance does and feels, that same meaning calls into play the listener's experience.

From this perspective, it is thus obvious that language can only be processed in the context of the discourse.[18] Following Benveniste (1966, 1974), by discourse we mean that every utterance presupposes a speaker and a hearer, and the intention in the former to influence the latter in some way. Indeed, the sentence only acquires reference as a function of the context in a given discourse – hence as a function of an agreement between speaker and hearer as to what is to be understood. Such reference pertains to the situation with which the discourse is actually dealing. It is in the context of the current discourse situation that the speaker refers to this or that entity through the use of language. This use combines the sense of the sentence (what the sentence means: reference to the world, predicating something) with a reference to the speaker (what the speaker means). In this way, predicate and logical subject are combined into one unified sentence.

In the discourse, the speaker takes up the position of enunciating subject. He can do so exclusively because he is able to say 'I'. This is possible only through the use of language. It is only in the current discourse situation that the speaker can appropriate the use of the personal pronoun 'I' and have it refer to himself: he is speaking to a 'you', the latter also being capable of saying 'I'.[19]

This is what Benveniste means when he says that in identifying himself as a unique person, the speaker who says 'I' takes upon himself the entire language. 'When an individual appropriates the language to himself, language is transformed into discourse situations, characterized by a system of internal references the key to which is "I", and which define the individual through the special linguistic construction of which he avails himself when he enunciates himself as speaker' (Benveniste, 1966: 305). In language use, then, 'I' alludes each and every time to the person whose experience is being referred to by the sense of the statement uttered.[20]

But what is the origin of the possibility of saying 'I'?

The assertion 'Let's go to the park', with which the caregiver interpellates the child, creating, through the use of language, a reference to the generic experience of that event, may be met by the child with a response conveying disapproval, such as 'I don't want to go to the park'. The I in 'I don't want . . .' indicates a position in relation to a you which is based on the being mine of experience. It is because ipseity experiences itself on each and every occasion as the source of a possible range of actions and emotions that going to the park as a possible mode of conduct of the experiential Self acquires significance. Saying 'I don't want to . . .' is founded on a practical dimension corresponding to the way ipseity understands itself when it is interpellated in the course of the discourse with regard to that possible situation. Seen in this light, meanings are possible ways of being.

But if saying 'I' is structured on the basis of the being mine of experience,[21] and if being mine is declined by way of each and every event, in order for it to be possible to say 'I', a sense of the permanence of the presence of the experiential Self over time must

already be organized in the antepredicative dimension. Obviously, this sameness of Self is realized at each and every event. This means that in the antepredicative dimension, the oneself which perceives itself and acts on each and every occasion must find itself each and every time possessing a sense of being the same.[22] This is the perspective implied by those studies in developmental psychology which stress the fact that a basic sense of Self is established in the newborn – well before the onset of mirror self-recognition – through the sensory-motor system, which enables the infant to locate his body with respect to the environment (Butterworth, 1990; Meltzoff, 1990).

The onset of this nonsubstantial permanence of Self,[23] whose specific feature is that of discontinuity, occurs at the age of 18–24 months, both in the domain of the senses and in that of language. When the child looks at himself in the mirror, he recognizes his mirror reflection as identical to himself. In other words, he can understand that, in each and every one of the different situations he encounters in his daily life, he is both different and concurrently the same.[24]

This capacity is also exhibited at the level of language with equally great force. It is what the use of language bears witness to. My experience, which is recounted in the first person, concerns my perceiving myself in this or that situation, and it corresponds contemporaneously to how sameness manifests itself on each and every occasion. Ipseity, which experiences this or that state of affairs while its continuity remains unaltered, is reflected and reconfigured in language through the 'I' which recognizes and talks about a particular episode, a given event. It is thus not attributable to chance that mirror self-recognition is related to the acquisition of personal pronoun use and pretend play, such that, during the second year, children showing mirror self-recognition used more personal pronouns and demonstrated more advanced pretend play than did children not showing self-recognition (Lewis and Ramsay, 2004; Courage, Edison and Howe, 2004).

Why does this ability emerge only at age 18–24 months? What is the relationship between the capacity to say 'I' and the ability to recognize the reflection in the mirror as one's own self-image?

2.8 Recognition of self in the mirror and in language

There is a story (Gallup, 1994; Povinelli, 2000), which it is said goes the rounds among those that work with the three species of anthropoid apes –the gorilla, the chimpanzee and the orangutan – that you can test the intelligence of these primates by leaving a screwdriver in the cage of each animal.

The gorilla looks at it, smells it and then ignores it. The chimpanzee jumps on it, puts it in its mouth and plays with it. The orangutan looks at it, pretends there is nothing there, takes it, hides it and that very night takes the cage apart.

What does this anecdote tell us? Obviously, the three species have different ways of using tools. We also know that while the chimpanzee and the orangutan exhibit a clear ability of self-recognition in the mirror, the gorilla is incapable of such an act. What is the relationship between self-recognition and the ability to use tools?

Povinelli (2000; Povinelli and Cant, 1995; Povinelli and Prince, 1998; see also Barth, Povinelli and Cant, 2004) explains this difference both from the standpoint of the real life of these primates and from the phylogenetic perspective. While the

orangutan, which is probably the nearest to our common ancestors, has a lifestyle that is totally arboreal, the chimpanzee lives both on the ground and in trees, and the gorilla has an 'earthly' lifestyle. Unlike other primates, which have stereotyped movements, the orangutan and the chimpanzee manifest a wide range of movements in the trees. In particular, given the relationship between bodily weight (40–80 kg) and the fragility of the branches along which it moves, the orangutan employs a type of movement called 'clambering', which combines the vertical position of the trunk with the use of various supports that it can grab hold of, depending on the type of movement to be executed. Since the orangutan lives several metres above ground level, it cannot afford to fall. Consequently, this characteristic type of movement obliges it to be constantly aware of its actions and its posture. In Povinelli and Prince's (1998) view, this capacity developed in the course of the Miocene age from an ancestor common to humans and to higher primates in order to maintain adaptation to an arboreal lifestyle in relation to a significant increase in bodily weight. The on-line sense of the position and of the movement of one's body is claimed to explain why self-recognition in the mirror is limited to the orangutan and the chimpanzee,[25] since these species still lead a life which requires a system of 'kinaesthetic awareness'. Thus, while these primates look at themselves in the mirror, they establish a relationship of equivalence between the behaviours which they observe of themselves in the mirror and the on-line experience they have of their own daily experiences.

The loss of, or failure to acquire, that capacity in gorillas, due to a difference in their evolutionary history based on ground locomotion, is claimed to explain, similarly, why this species cannot recognize itself in the mirror. According to this interpretation, it should be clear why the more 'arboreal' species exhibit greater ability in manipulating tools. This type of existence is claimed to favour the development of more precise control over behaviour, hence of a more sophisticated body schema.

The hypothesis therefore follows that, in order to maintain their adaptation to a highly variable arboreal environment, which requires frequent evaluation of new information (Povinelli and Cant, 1995), the orangutan, and to a lesser degree the chimpanzee,[26] has developed a more sophisticated visuo-tactile-motor system to manage its actions more effectively and more efficiently. Such a system would therefore ensure a more successful integration of body and peripersonal space in an ongoing global sense of Self. This would seem to indicate that a more sophisticated mode of executing actions corresponds to greater and improved mastery over one's body and a more integrated and clearly-defined body schema. Continuing to follow this hypothesis – albeit in a different direction from that taken by Povinelli – it would appear to be possible that, in anthropoid apes, the MNS is part of the mechanism underlying the workings of a sort of 'matching system' between the potential range of actions and the action actually carried out, the goal of the system being to favour greater motor control in an environment characterized by great flexibility.[27] This could then account for the capacity that these primates exhibit to form a relationship of 'motor equivalence' between the behaviours reflected in the mirror and the actions they carry out, so much so that, after a certain period of exposure to the use of the mirror, they actually begin to recognize their own reflected image. Moreover, the refining of the ability to execute actions could have indirectly aided the ability to recognize congruence between the acts these apes see others perform and their own behaviour, thereby improving the capacity to predict the consequences of the action of their conspecifics.

This hypothesis, which accounts for the necessary connection between the ability of mirror self-recognition and the development of a unitary visual-proprioceptive sense of one's body, might also explain the difference between higher primates and Homo sapiens. Indeed, exactly the same mechanism employed by anthropoid apes to match kinaesthetic and visual information could have been exploited anew (through the mechanism of exaptation) in the course of human evolution as the neural substratum to recognize oneself as an 'I', in the first person, both visually and through language (*viz*. acoustically).

Unlike the great apes, however, in humans it is obviously the face that plays the most significant role in the recognition of Self and in the comprehension of the actions and intentions of others.

In addition to being at the centre of the world of human emotions, it is also, so to speak, the site of language. As is suggested by many studies on the relationship between audiovisual speech perception and the motor activation of areas involved in planning and executing speech production, the dynamics of facial movement and speech perception-production are clearly integrated (Skipper, Nusbaum and Small, 2005; Hall, Fussell and Summerfield, 2005; Pulvermüller *et al.*, 2006).

From an evolutionary standpoint, the social centrality of the face runs parallel to the emergence of more complex intersubjective dimensions (e.g. closer caregivers–infant relationships, bonds with lovers, friends, long-term mates, family members), implying a more sophisticated ability to tune into the face of individual others (Fridlund, 1994; Parkinson, Fischer and Manstead, 2005; Pulvermüller *et al.*, 2006). Areas within the inferior occipital-temporal region have in fact been identified that are selectively involved in the perception of unknown faces (Kanwisher, McDermott and Chun, 1997; McCarthy *et al.*, 1997), while other areas identified are specifically activated by personally familiar faces, and still others are associated with self-face recognition (Platek *et al.*, 2006).

If in humans the face plays the most important part in recognizing the Self, this difference from anthropoid apes would lead us to infer, from an evolutionary standpoint, that self-face recognition involves the same observation/execution action-matching mechanism recruited in primates for recognizing their own actions and those of others. Three recent fMRI studies on self-face recognition have shown that recognizing the image of one's own face is accompanied by activation of the motor system (Platek *et al.*, 2006; Sugiura *et al.*, 2005; Uddin *et al.*, 2005; Uddin *et al.*, 2007). In other words, in humans those very same mechanisms underlying 'kinaesthetic awareness' in primates can support face-representation functions; they can map the image of one's own face on to one's own motor system. These studies indicate that the right frontoparietal areas[28] – which are typically activated when participants are presented with the image of their own face – overlap with areas that belong to the MNS, that is, they belong to a system employed to map perceived action on to one's own motor system. From this perspective, we can readily comprehend why toddlers who pass the mark-test of mirror self-recognition have an equivalence in performance on the task of recognizing their own leg (Nielsen, Suddendorf and Slaughter, 2006). Moreover, confirmation of this type of overlap may be had from studies on the development of body self-awareness in toddlers (Brownell, Zerwas and Geetha, 2007; Moore *et al.* 2007).

What is the relationship between the ability to identify one's own image in the mirror and the capacity to say 'I'?

Obviously, the fact that the child begins to deploy language through the use of phrases during more or less the same period in which he acquires the capacity to recognize himself in the mirror (through recognizing his own facial expressions) is not accidental. Up to that point, he could only reflect himself in the behaviour of others, despite the fact that he had developed a sense of the permanence of Self. Put differently, despite the fact that he was able to perceive uninterruptedly the sense of the being mine of experience, he recognized its meaning only through reflection in shared behaviour.

The possibility of reflecting himself in his own actions and emotions, of grasping himself as identical to himself (who feels and acts), generates a new feeling of intimacy with himself and of demarcation from others. The child begins to develop a sense of the privacy of experience – which does not need to be interior – a clear indicator of this development being the monologues with himself.[29] This is accompanied by a sense of the 'property' (*Eigenschaft*) of his modes of being, which is discernible in his change of attitude toward himself and toward others. This reflection without interiority hides a new tension: the reconfiguration of Self enabled by the onset of language use defines not an egoic Self or the closure into the Self, but determines the way in which ipseity becomes its own.

Before we can proceed in our discussion in the direction indicated by the question of 'who?', we must first make a small digression to deal with a topic which psychology has linked closely to the subject of the recognition of the Self in the last few years, namely, self-conscious emotions. A series of observations appear to suggest that these emotions already emerge before the development of reflective consciousness.

2.9 Affective engagements

There is a great deal of literature, above all of a representational nature (Lewis, 1987, 1992a, 1992b, 1993a, 1993b, 1995), which underscores the point that during the child's second year of life there emerge new emotions – the self-conscious emotions – whose genesis is ascribed to a changed level of awareness of Self and of others. The nature of such emotions is hypothesized as being characterized by the fact that they develop in the context of a new relational state: that in which the child is self-conscious. These emotions (pride, embarrassment, shame, guilt, envy) would appear to take shape through the conscious evaluation the child makes of his behaviour in relation to the reference parameters.

Other studies which do not concur with this approach emphasize the fact that interpersonal emotions (pride, embarrassment, shame, guilt, envy, coyness, shyness, teasing, humour, emphatic concern and so forth) were already present in the first few months of life. As Vasudevi Reddy (2003) forcefully points out, if the infant is the focus of the caregiver's attention then the relation between the observing parent and the observed child can be experienced directly by the infant; it does not need to be inferred, as the representational hypothesis argues. Awareness of the attention of the other on the oneself corresponds to a mode of self-perception, an affective consciousness of Self. The link between how the caregiver's attending is perceived and how one's own ipseity is experienced – which is different and specific to each and every one of the self-conscious emotions – is directly felt right from the earliest months of the child's life. How the infant perceives the other's gaze depends not on 'beliefs about others' evaluations', but on how he feels at that moment.

For example, as is reported in the study quoted earlier by Meltzoff (2007), after having observed the irritated reaction of an adult on seeing a third party manipulate new objects, the child imitates his behaviour only if the adult's back is turned. Thus, the infant has a keen awareness of the standard and of the rules that govern a given situation without his having to possess an internal representation of himself, of the other, and of the Self as perceived by the observer (Trevarthen, 1992; Draghi-Lorenz, Reddy and Costall, 2001; Reddy, 2005). It is no intellectual feat to predict that were the adult to suddenly turn round and so catch the child 'with his hand in the cookie jar', the child would be embarrassed. In other words, perceiving adult attention on the child refers the child back to the awareness of the (emotional) significance that the caregiver attributes to the child's behaviour and, at the same time, to the inadequacy of that behaviour in the adult's presence.

The child's embarrassment reflects the incompatibility between the adult's expectations, which the infant has anticipated, and the real ongoing situation. It is this incongruence (which corresponds to being found out) which creates a greater focus on oneself, precisely because the child's conduct becomes the focus of attention of both interactants – caregiver and child.[30] Since embarrassment conveys discomfiture with regard to the current situation, it opens up the possibility of repair (Semin and Manstead, 1982). This type of embarrassment is closer to shame.

In any case, even when embarrassment is viewed as a form of shyness, hence not as a result of having committed a *faux pas*, as, for instance, when children are asked to perform, the focus on oneself is retained by the awareness of being at the centre of the unwanted attention of others.

The interpersonal – rather than the representational – origin of these emotions seems to apply to other social emotions such as jealousy, envy, pride, guilt. Hart (Hart and Carrington, 2002; Hart *et al.*, 1998), for example, showed in two studies how six- and twelve-month-old infants manifested distress – interpreted as jealousy – when their caregivers paid attention to a lifelike baby doll. Draghi-Lorenz, Reddy and Costall (2001) report the observations of parents who described a form of distress – interpreted as envy – when their children wanted something (e.g. a toy) possessed by another infant. Pride may be considered to be an emotion which emerges in concomitance with the positive attention (praise) received following a particular achievement (involving a feeling of mastery), rather than being dependent on beliefs about oneself as worthy (Draghi-Lorenz, Reddy and Costall, 2001). Finally, guilt could emerge as a direct perception of the damage caused by an action produced by the child and detrimental to another child who manifests grief at the wrong done to him, rather than as a result of the appreciation of self-blame in a reflexive space. Indirect confirmation of this hypothesis comes from a recent study, which shows that six- and ten-month-old infants take into account an individual's actions toward others in evaluating that individual as appealing or aversive (Hamlin, Wynn and Bloom, 2007).

Not only is each of these social emotions generated in response to a determinate ongoing situation, but in its turn it opens up new possibilities of actions and relationships. Jealousy is a spur to the reestablishment of a privileged situation, envy redirects the caregiver's attention to a desired object, pride creates an effect on the audience, guilt paves the way to repair (Parkinson, Fischer and Manstead, 2005). The occurrence of these emotions does not involve conceptual thought – rather it lays the foundations for this possibility.

It is obvious from this perspective that, rather than being at the root of self-conscious emotions, as is claimed by the representational standpoint, reflexive consciousness,

or more accurately, the way in which reflexive consciousness takes shape, is strongly influenced by how affective interpersonal space has been constituted in the preceding months. The fact, that is, that the child is the centre of attention of others, and that this convergence of social attention creates in him social emotions which position him in interpersonal space, may be hypothesized as constituting the necessary condition for the development of reflexive capacities, rather than being the result of such capacities. As Heidegger (1988) stated (in reply to Natorp, who emphasized the distorting nature of reflexive consciousness), reflecting is only one way of grasping oneself, one way that enables comprehension to intensify, rather than constituting the foundational base of self-comprehension.[31]

2.10 Acting and speaking

What, therefore, is the relationship between the ability to identify one's own image in a mirror and the capacity to say 'I'?

From the beginning of his third year a child is able to recognize that he is identical to himself, namely, that he is one and the same Self in each and every situation, both in his actions and in his language. Thus, while he looks at his face reflected in the mirror, he identifies it as his own. In the same way, he identifies himself linguistically as the subject of a statement, that is, he acquires the capacity to grasp each and every time what the sentence means (shared sense), referring it to his own experiential perspective. Thus, he establishes a relationship of equivalence between experience referred to by the sense of the sentence and his own experience. In this way, he appropriates the sense (which the sentence refers back to) while concurrently recognizing and articulating his experience, which is reflected in the conceptual content of the sentence, and which that content interpellates.

Whereas mutual comprehension was achieved, in the previous stages, through (emotionally situated) action, it now begins to be mediated by language. As is shown by Emily's monologues studied by Nelson (1996), the contents of these first rudimentary ventures into storytelling are real-life experiences, actions and passions, all of which recount daily existence. The (shared) sense of actions and passions is reinscribed in the sentence (Engel, 1986; Dunn, 1988). The structure of action and the structure of language gradually come to be superimposed on one another, inasmuch as both are characterized by a shared sense and contemporaneously by the reference to oneself.

From a neuroscientific standpoint, the hypothesis gains further confirmation from an examination of the organization of Broca's area, formerly considered exclusively a speech production area, but where, it has now been discovered, we also find neurons that are activated both by orofacial acts and by hand gestures. It is interesting to note that, in addition to playing a crucial role in language production, Broca's area also plays an important part in imitation and in action recognition (Binkofski *et al.*, 1999; Iacoboni *et al.*, 1999; Molnar-Szakacs *et al.*, 2002; Iacoboni *et al.*, 2005). This region is known, in fact, to be part of the MNS (Bookheimer, 2002; Rizzolatti and Craighero, 2004; Nishitani *et al.*, 2005; Gallese, 2005). Several works by Gentilucci and his colleagues (Gentilucci, 2003; Gentilucci *et al.*, 2001, 2004a, 2004b, 2006; Bernardis and Gentilucci, 2006) have found a close relationship between speech production and the observation/execution of arm and hand gestures, thereby corroborating the relationship between manual and oral language-related gestures.

Thus, on the one hand, the meaning of a gesture is fixed in sedimented sense structures: intentional content[32] – it is common sense that renders actions familiar and actions shared, such as routines or games. On the other hand, meaning refers to the real experience of a 'who' at a given moment and in a specific situation[33] – this is the meaning that the action has for the person carrying out that very action.

In the same way, the meaning of the sentence (that is, its propositional content), on the one hand, is held constant by what the sentence refers to, and, on the other hand, meaning brings to language the speaker's experience. It is precisely because the speaker and the listener can share sense content that one can convey one's experience to another. It is the sense content that, realized on each occasion by each individual speaker through concrete reference to his own experiential domain, enables us to include and grasp the experience being communicated. 'Communication can succeed,' write Liberman and Whalen (2000), 'only if the two parties have a common understanding about what counts – what counts for the speaker must count for the listener'.

How is this equality concerning 'what counts in communication' established?

It is only possible because there exists, in humans, an echo-neuron system, that is, a system that matches verbal perception with speech production (Rizzolatti and Craighero, 2004).

On the one hand, this system can mediate the imitation of verbal sounds, as in the case of the child repeating words which do not refer back to any of his experiences and which are therefore meaningless. For instance, Levy and Nelson (1994) investigated children's acquisition of causal and temporal words ('because', 'yesterday', 'soon' and so on) and found that children's production of those linguistic items was bound to specific discourse contexts and realized exactly the same discourse functions that their parents used them for, without possessing fully adult-like understanding and flexibility of use. (Any adult learning a second language finds himself in a similar situation, that is, of learning a series of words without understanding what they mean).

On the other hand, this echo-neuron system may also perform a semantic function inasmuch as it enables the hearer to refer the words employed by the speaker to his own experiential domain.

What role does the echo-neuron system play in the process of the meaning of an action being conveyed by the sound of a sentence?

For meanings to end up in words, the articulation of sounds must be bonded with the sense of an action. In other words, there must exist shared neural mechanisms capable of linking up action observation and understanding[34] to language production–perception, hence, a neural substratum that is capable of relating acoustic perception to linguistic production and to a shared sense of action. A number of recent studies indicate that the MNS plays a key role in the articulation of the complex relationship between action, language and communication.

In the course of development – from the child's very first words, but even more clearly with the onset of language – the child gradually learns that during communication the sounds, the words which accompany experience refer to that specific current experience itself, to the point that the child also learns to predict what will happen next. This gradual transformation in the way the child communicates is fostered by the caregivers, who generally grasp, underline and repeat those words most relevant for each individual communicative context. This enables the child to integrate the words that are used and repeated during the execution of shared activities into his pre-reflexive comprehension

of experience. 'The early establishment of shared meanings,' writes Nelson (1993: 116), 'thus involves an interaction of the child's preverbal interpretation of events and objects, and the adult's use of those words, and acceptance of the child's uses within jointly constructed and mutually understood activities'. These observations suggest that speech perception probably shares a common neural substrate not only with the regions assigned to (a) speech production – because of the parity between sender and receiver of the message invoked by Liberman – but also with the areas engaged by (b) perception – the execution of action.

With regard to the first point (language production–perception), various studies highlight the role of motor areas capable of *resonating*, of activating themselves, by listening to speech. Two transcranial magnetic stimulation (TMS) studies have reported facilitation of tongue and lip muscles when subjects listened to speech (Fadiga *et al.*, 2002; Watkins *et al.*, 2003). A third study (Meister *et al.*, 2007) showed that the disruption of human premotor cortex impairs speech perception. Other fMRI studies reported (a) the activation of motor areas involved in speech production in passively listening to meaningless monosyllables (Wilson *et al.*, 2004), (b) the recruitment during speech perception of specific motor circuits that reflect distinctive phonetic features of the speech sounds encountered (Pulvermüller *et al.*, 2006), (c) an increase in motor system activation on hearing unfamiliar non-native phenomena, compared to that during the perception of native phenomena (Wilson and Iacoboni, 2006).

The second point, which concerns the relationship between action and language, has been developed in a series of studies on the comprehension of action-related sentences aimed at uncovering, above all, the relationship between action-related sentences and the activation of motor areas related to the actions that one can perform oneself (and to which the senses of those sentences refer) (Buccino *et al.*, 2001, 2005; Hauk, Johnsrude and Pulverm¸ller, 2004; Tettamanti *et al.*, 2005; Gazzola, Aziz-Zadeh and Keysers, 2006). The most recent study in this field, which was carried out by Aziz-Zadeh *et al.* (2006) and which replicates and expands on some of the previous studies, shows that reading phrases describing actions carried out with different effectors (hand, mouth, leg) led to activation of specific motor and premotor areas that are active when an individual observes the same visually-presented actions.

This equivalence in human premotor cortex between areas that respond to both the execution and the observation of actions and areas activated by exposure to their verbal description, indicates that sense is grounded in action and passion, that language is rooted in one's experience of being-in-the-world. It might be said that discourse reconfigures what happens in the domain of acting and feeling in the domain of language, and it is precisely on the basis of this possibility of referring to the world in one way or another that discourse is regulated by each and every individual.[35] It composes what is pre-reflexively manifest but, at the same time, it can do so because it stems from, so to speak, communication with others, and via this channel, from the cultural community to which the communicants belong.

As we have seen, alterity intertwines with ipseity a first time at the prereflexive level; it becomes part of the domain of the meaning of subjectivity a second time simply through the use of language. It is in the reception of a discourse that is already under way, a language that has already been instituted, and in having been initiated into a cultural and historical world that is already there, that the shared sense structures are supplied, enabling one to grasp and recognize one's own experience.

Appropriating shared sense thus corresponds for everyone to the possibility of understanding oneself in this or that manner in the various ongoing situations and thus to *communicate* one's own experience of existing. 'Just as our common belonging to the same world presupposes that my experience, as my own and original experience, is the experience of being,' wrote Merleau Ponty (1969: 194), 'similarly, our belonging to a common language, or even to the common universe of language, presupposes a primordial relationship of me to my word, which bestows on it the value of a dimension of being, in which X may participate'.

But it is precisely language that guarantees the possibility of reconfiguring the experience of living in a unitary connectedness exhibiting the features of narration. And this throws new light on the relationship between agent and action. The action, for instance, of Mary taking hold of the cup, considered from the standpoint of the narration of Self, is influenced by new determinations. The agent–action relationship is no longer treated as if the agent were reabsorbed by the goal of the action, as if he were a function of that particular action. Instead, despite the fact that individual action has common-sense meaning for everyone, such action, which is reconfigured in language, takes on a meaning that is personal by dint of the fact that it is found within a configuring framework, the narration of oneself. Individual action is placed within a referential context by the story that it helps to articulate and develop, both in the case where the event falls within the range of expectations that derive 'naturally' from a narrative, and in the case where it takes 'by surprise' the predictions made regarding the development of the story, thereby generating new expectations. From this standpoint, assimilating experience, hence historicizing it, does not mean instituting a chronological order, simply putting events into a sequence. What must be considered is the importance of a given experience in relation to the economy of the narrative, which confers upon the experience a value that changes with the changing of the story itself. That is to say, the meaning of acting and of feeling goes well beyond its empirical dimension.

Every human being integrates the events he experiences to combine the present with sedimented experience, and with the horizons of his expectations, all at the same time. The special relationship between these different components is realized in the continual reshaping of the narrative of oneself (in its various forms, including those of the fragmentation of the story). The narrative recomposition of the experience of living reconfigures the variety of one's own experience into a signifying totality, while concurrently delineating the person to whom those actions and emotions refer. After Ricoeur, we turn to narrative identity to indicate this dynamic modality of composing personal identity.

Endnotes

[1] Merleau-Ponty (1962: 493).

[2] The empirical tools for this new issue came from the development of the first false belief task, which was to remain paradigmatic, by Wimmer and Perner (1983).

[3] Theory of mind mechanisms.

[4] Theory theory.

[5] In a recent study, Fogassi *et al.* (2005) show that in the inferior parietal lobe of the ape, motor neurons which codify specific acts manifest a different mode of activation when those acts are part of different actions (contexts).

[6] Structuring action in linguistic terms is reflected in the syntactic structure of the clause, as realized by the predicate (the propositional content) and the subject (the identifying referent).

[7] With regard to this topic, a brief analysis of the original meaning of *similis*, *-e* is of interest. According to Semerano (1994), right from the archaic form *semol*, *semul*, which then produced the adverbial form *simul*, this lexical item indicated the concatenation of two simultaneous actions which took place at the same time. Semerano also holds that *similis* expresses only similarity and is originally a calque from *omalos*, united, and on *simul*, whose original meaning is 'united with', 'together with'. The term 'simulation' could thus indicate an action which co-occurs, at the same time as another action, and which is similar to that action.

[8] We grasp the other in his worrying, in his hurrying, in his being disgusted, in his going to work – but what escapes us? What residue is left out of our comprehending the other? What do we fail to consider? The face, says Levinas (1961), to indicate that absolute manifestation of the other, irreducible and unknowable; the face, a trace of infinity.

[9] It is the fact of being born that founds the possibility of initiative, the capacity of giving life to a new action. According to Hannah Arendt (Arendt, 1998: 182), 'the unprecedented originality . . . [of Jesus of Nazareth's comprehension of the capacity to act] can be compared to the Socratic comprehension of the possibilities of thought'.

[10] The ontological role of proto-linguistic communication is also underscored by Bateson, who had the first intuitions which gave rise to the study of this domain. She writes (Bateson, 1979: 74), 'Proto-conversations tend to take place after feeding, caretaking, bathing, dressing – at a moment when the bustle is over and physical needs are met and sleep is not urgent. These are the moments that in many households in our culture are kept almost ritualized for the most intimate converse of parent and child.'

[11] With regard to this point, we could take up Winnicot's (1965: 145) definition of a 'good enough' mother as someone who meets 'the omnipotence of the infant and to some extent makes sense of it'.

[12] Non-basic emotions may be defined as those emotional responses which are not attributable to the activation of affect programmes understood as complex, coordinated and automated sets of bodily changes.

[13] Emotion is the subject of Chapter 4.

[14] This domain is, as yet, virgin territory, no research having been done in the area.

[15] With regard to the close connection of the emergence of words to the context, it is significant that 'disappearance' words (e.g. 'gone') emerge in relation to the capacity that the child develops of looking for a previously visible object which has been placed out of sight (Gopnik, 1984).

[16] Some infants start to show evidence of mirror self-recognition from just 1.5 years old, and by the age of 2, about half of all infants demonstrate that they have acquired the skill.

[17] Language becomes symbolic. *Symbolon* means keeping something together using something else. In a symbol, man keeps himself united with something else (the world in which he lives) and at the same time he comes to an agreement with other men with regard to how that symbol should be interpreted. This allows Heidegger (2001a: 309) to state: 'Words emerge from that *essential agreement* of the human beings with one another, in accordance with which they *are open in their being with one another for the beings around them*, which they can then individually agree about – and this also means fail to agree about. Only on the grounds of this originary, essential agreement is discourse possible in its essential function: [*semainein*], giving that which is understandable to be understood'.

[18] The linguistics of the sentence give precedence to semantics inasmuch as they are immediately concerned with the concept of sense. Thus, Ricoeur (1976: 21) writes, 'the most concrete definition of semantics is the theory that relates the inner or immanent constitution of the sense to the outer or transcendent intention of the reference'. From this perspective, grammar can only derive from discourse.

Deriving grammar and semantics from discourse is underscored in the domain of developmental psychology by studies carried out by Nelson (1986; 1996: 144) 'Discourse is the source of knowledge of language forms and is the context within which meaning and use of language forms is developed and elaborated'.

The results of a recent MRI study (Brauer and Friederici, 2007) on 5–6-year-old children investigating the semantic and syntactic aspects of sentence processing found a large activation overlap for these two language aspects in the superior temporal gyrus (STG) in contrast to adults who exhibit function-specific activation. Such results support the hypothesis that there exists a common neural substratum which underpins the development of semantics and syntax.

[19] This 'I–you' opposition, which lies at the heart of the linguistic theory of person, emerges at about 18–20 months of age, during which time the infant begins to mark a person contrast in some explicit way. Bates (1990) calls this new phase 'contrastive reference'. The linguistic category of 'person' belongs exclusively to the position 'I' and 'you'. 'I' designates the speaker, who is also the person the utterance refers to, while 'you' refers to the hearer, the person whom the 'I' is addressing by means of his current utterance. The third-person 'he', 'she', 'it' is not a specific person or entity – it is impersonal, such that a *thing* is predicated in the third person. 'In actual fact, one of the features of the persons "I" and "you" is their specific uniqueness: both the enunciating "I" and the "you" addressed by the "I" are each and every time unique. But "he-it" may be an infinity of subjects – or no-one?' (Benveniste, 1966: 275).

[20] As Ricoeur (1976: 20) says, 'the sense is traversed by the referring intention of the speaker'.

[21] 'Because Dasein has *in each case mineness [Jemeinigkeit]*, one must always use a *personal* pronoun when one addresses it: "I am", "you are"' (Heidegger, 1962: Section 9, page 68).

[22] From this standpoint, time becomes the principle on which identification is based, inasmuch as that which is my own is assigned to me on each and every occasion.

[23] That is, not connected to a consideration of the mind as immaterial substance.

[24] The perspective being argued here is clearly different from Gallup's (1982) viewpoint, summarized by Platek, Myers and Gallup when they write 'to recognize your own face in the mirror, you have to first have a concept of self (i.e. you have to know who you are to be able to recognize who is being reflected in the mirror). Therefore, the ability to recognize oneself in the mirror appears to represent a cognitive capacity related to a sense of self-awareness' (Platek, Myers and Gallup, 2004: 116). In actual fact, the identification of a mirror reflection as one's own is not a cognitive task dependent on pre-existing self-knowledge. The emphasis we place on the expression 'each and every time' underscores the fact that a relationship of equivalence is set up between the behaviour observed in the mirror and one's own experience every time the event occurs, without requiring conceptualization or representation.

[25] There is strong evidence that the bonobo (or pygmy chimpanzee) is also capable of mirror self-recognition (Hyatt and Hopkins, 1994).

[26] Since it leads a ground–tree existence.

[27] That is, control over actions extended to the domain of potentiality, induced by the specific quality and the spatial location of the targets of such actions (Gallese, personal communication).

[28] Hemispheric lateralization in prefrontal cortex during self-face recognition is still a source of debate (Keenan *et al.*, 2000, 2001, 2003; Turk *et al.*, 2002, 2003).

[29] In his 'inner' speech the child creates a dimension of closure between oneself and himself through changing the roles of both speaker and hearer. For an investigation of crib speech, see Halliday (1975) and Nelson (2006).

[30] This cannot happen in younger infants, because they are incapable of recognizing themselves in their own action and emotions, while they can, however, reflect themselves in those of others. This is why embarrassing situations are characterized by a greater focus on the other (Reddy, 2003).

[31] New studies on the relationship between attention and emotional display in the course of mutual engagement might furnish interesting data on this topic.

[32] If we describe the intentional content of action in terms of a goal that has to be achieved, we can see the neural substratum that discharges during purposeful movements as a sort of motor-embodied vocabulary. This is how Rizzolatti and Gallese (1997) describe the set of different types of goal-directed neurons coding different motor acts related to prehension, discovered in area F5 of the chimpanzee's premotor cortex.

[33] We have seen the role played by the MNS in Mary's case in the construction of shared sense at the level of action.

[34] It should not be forgotten that observing a meaningful action is always already comprehending that action.

[35] 'Not only must a pre-predicative manifestness in general constantly already occur and have occurred, however, if the assertion as pointing out is to be accomplished in whatever way, but this *pre-predicative manifestness* must itself be this *occurrence* in which a particular *letting oneself be bound* occurs. This is the prior relation to that which gives pointing out in assertion its measure: beings as they are. . . . Letting oneself be bound must already bring toward itself in advance, and as something that can be binding, whatever is to provide the measure and be binding in one way or another' (Heidegger, 2001a: 342).